A Microservices-Based Home Energy Tracker for Event-Driven Monitoring, Deduplicated Alerting, and ML-Based Anomaly Detection

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Abstract - Residential energy platforms require a traceable route from meter-like telemetry to controlled feedback. This study implements a local 17-service Docker composition combining Wokwi-compatible ESP32 telemetry, API-gateway admission control, Apache Kafka, InfluxDB, MySQL, deterministic insights, and duplicate-aware alert delivery. User-facing routes apply Keycloak JWT, role, and ownership checks, while simulated-device ingestion uses a device-bound token. A baseline Random Forest artifact, random_forest-csv-20261003T205417Z, is selected by validation F1 from the supplied labelled CSV; its historical held-out precision, recall, F1, ROC-AUC, and PR-AUC are 0.880952, 0.925000, 0.902439, 0.997430, and 0.948532, respectively. On 2026-10-11, the live ML container reported the baseline model UP and its legacy Kafka worker connected; a Wokwi-compatible three-field input with energyConsumed = 0.55 returned anomaly = false, anomalyProbability = 0.0, decisionThreshold = 0.31, and four explicit contextual-feature imputations. The isolated REFIT controlled branch was disabled and contributed no live result. Historical controlled local checks recorded duplicate suppression within 15 min, an eight-target Prometheus snapshot, and 450/450 accepted k6 ingestion requests. The results demonstrate implementation-level integration and bounded local evidence, not physical-meter accuracy, field-model generalisation, production security, or production-scale resilience.

Keywords - *Home Energy Tracker, Internet of Things, microservices, Apache Kafka, anomaly detection, Keycloak, alert deduplication, Wokwi simulation.*

I. INTRODUCTION

Residential electricity bills provide limited visibility into when energy is consumed, which registered device contributes most, or when a configured threshold is exceeded. A practical Home Energy Management System therefore needs an auditable path from telemetry to stored records, interpretable outputs, and controlled notification. Recent IoT systems demonstrate affordable sensing and remote visualisation, whereas other studies focus on energy analytics or anomaly detection [3]-[6], [13], [16], [18]-[21], [23].

This work addresses the integration of a local, event-driven energy platform rather than the accuracy of a physical meter. The deployed composition separates user and device metadata, ingestion, usage, insight, alert, API gateway, and ML inference functions, and includes Kafka, InfluxDB, MySQL, Mailpit, Keycloak, Prometheus, Grafana, and a frontend. Wokwi supplies timestamped synthetic JSON payloads to the same HTTP ingestion contract intended for a future meter; Kafka decouples request acceptance from downstream persistence and analysis. The active ML component is a baseline FastAPI service and Kafka worker; the separately implemented REFIT controlled adapter is disabled in the current Compose configuration.

The study makes four bounded contributions: it implements a 17-service local Docker path from simulated telemetry to time-series storage and access-controlled APIs; it trains and deploys a fixed baseline anomaly-inference artifact while retaining deterministic bill, ranking, standby, and threshold functions; it applies device-bound telemetry admission, downstream caller checks, and duplicate-aware notification; and it records controlled local evidence for end-to-end behavior, gateway security, alert deduplication, observability, ingress load, selected outage conditions, and current baseline runtime status. The scope excludes calibrated sensing, utility-bill validation, physical-field generalisation, production penetration testing, and production-scale resilience claims.

II. RELATED WORK AND RESEARCH GAP

A. Existing Approaches

The literature represented in the source paper spans three complementary directions. Low-cost IoT systems use ESP32-class controllers and energy sensors to expose consumption or power-quality information [3]-[6], [13], [16], [18]-[21], [23]. Their measurement evidence cannot be transferred to the Wokwi-generated values used here. A second direction uses machine-learning or statistical methods for appliance classification, anomaly detection, and behavioural analysis [1], [7]-[11], [15], [22]. The present system evaluates a binary classifier only on its supplied labelled CSV and does not equate those results with real-household anomaly performance.

A third direction considers microservices, cloud organisation, or distributed control for energy-management systems [2], [12], [14], [17], [24]. These works motivate service separation, event processing, and maintainability, while leaving a practical need for a traceable prototype that links a telemetry contract, access-controlled API boundary, event broker, time-series persistence, deterministic insights, anomaly inference, and duplicate-aware notification. The present work adopts this integration focus without claiming equivalence to the scale or fault tolerance targeted by prior architectural research.

B. Research Gap and Contribution

The research gap is limited implementation-level linkage between residential telemetry, event-driven services, controlled access, historical storage, transparent analytics, anomaly inference, and alert delivery with explicit verification boundaries. The proposed system addresses that gap through Wokwi-compatible telemetry, device-bound gateway admission, Spring Boot microservices, Kafka, InfluxDB, MySQL, a deployed baseline Random Forest

artifact, deterministic insight functions, and Mailpit-backed alerting with a configured deduplication window. Remaining gaps include calibrated sensing, documented dataset provenance, per-household contextual labels, field validation, production-grade security assessment, and distributed resilience evaluation.

III. PROPOSED SYSTEM AND METHODOLOGY

A. Architecture and Telemetry Path

The runtime input contains deviceId, energyConsumed, and timestamp. Wokwi firmware emits this JSON payload at an approximately 8 s interval; Eco, Heavy, and Alert Test modes generate controlled values for integration demonstration and are not calibrated physical measurements. User-facing requests enter through the API Gateway with Keycloak JWT, realm-role, and ownership checks. The gateway replaces trusted identity headers, and Wokwi-compatible ingestion requires a device-bound X-Device-Token; downstream calls use separate static gateway or background-service credentials. The Ingestion Service validates accepted payloads and publishes energy-usage events to Kafka.

The Usage Service and the FastAPI ML service independently consume energy-usage events. Usage writes time-stamped energy_usage points to InfluxDB and derives selected-window aggregates. The ML service hosts KafkaInferenceWorker, which invokes the fixed baseline Random Forest artifact random_forest-csv-20261003T205417Z and publishes anomaly-results; repeated processing failure is routed to anomaly-processing-failures before offset commitment. For the three-field Wokwi payload, Temperature, Humidity, Wind_Speed, and Avg_Past_Consumption are median-imputed and disclosed in the prediction response. User and Device Services maintain household, appliance, and alert-setting metadata in MySQL. Threshold and anomaly paths reach the Alert Service, which persists notifications and uses a configured 15-min deduplication window. Prometheus and Grafana provide service-level target and metric observation. The isolated REFIT controlled route is disabled and is not connected to Wokwi energy-usage or legacy alert topics.

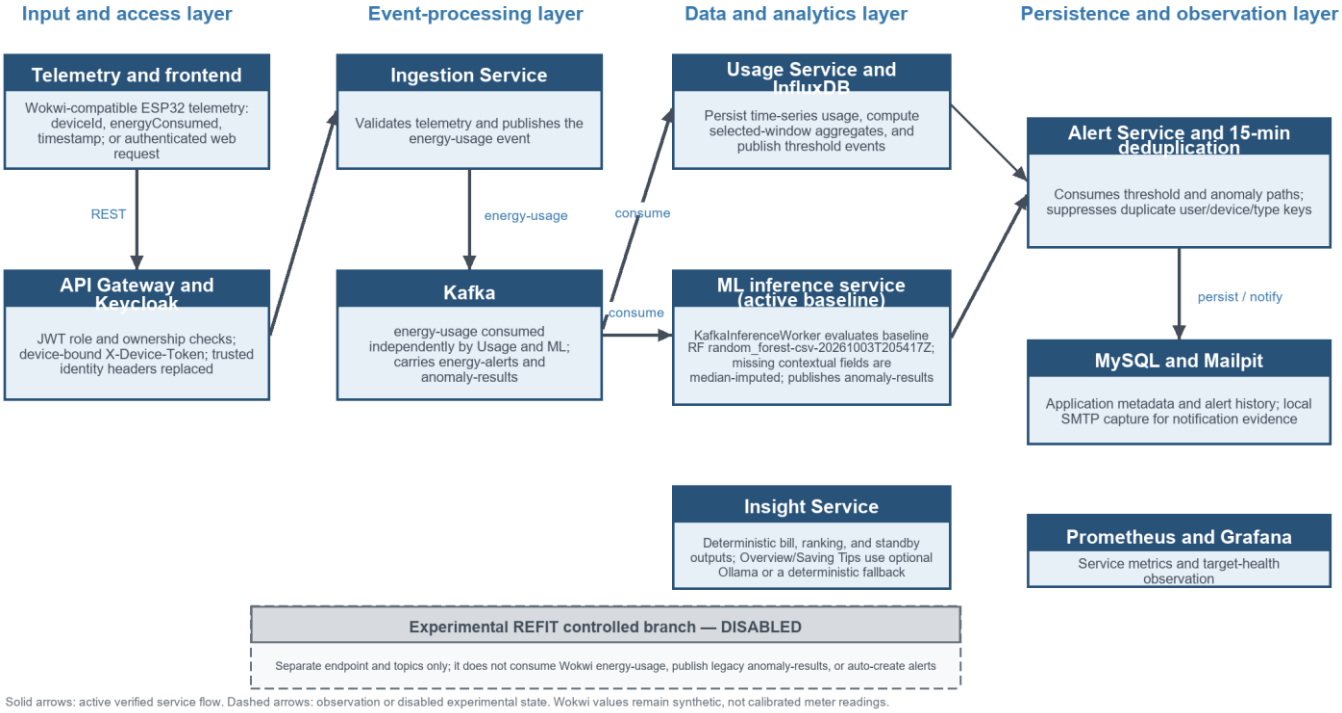


Fig. 1. Verified service architecture, active baseline inference path, and isolated disabled REFIT branch of the Home Energy Tracker.

B. Deterministic Insight and Alert Methods

The Insight Service derives deterministic outputs from aggregated usage. It computes a daily rate and a 30-day energy projection from total energy E over D selected days. The configured local tariff uses a fixed 10.00 charge and slab rates of 0.12 per kWh for 0-100 kWh, 0.18 per kWh for 100-300 kWh, and 0.25 per kWh above 300 kWh. Device ranking uses each device total divided by the household total, and the standby estimate remains 20% of average load. Overview and Saving Tips may request optional Ollama text enrichment, but the verified 2026-10-11 response used deterministic fallback (isAiGenerated = false). These are demonstration rules, not a utility tariff, an appliance-level standby measurement, or a verified live generative-AI result.

$$E_{day} = \frac{E_{tot}}{D}; \quad E_{30} = 30\ x\ E_{day}$$

The anomaly branch uses the supplied smart_meter_data.csv with 5,000 records and seven columns, including Electricity_Consumed, weather/context fields, Avg_Past_Consumption, and Anomaly_Label (4,750 Normal and 250 Abnormal). Records are stably ordered by timestamp; Hour, DayOfWeek, and ConsumptionDeviation = Electricity_Consumed - Avg_Past_Consumption are derived. A chronological 3,500/750/750 train/validation/test split compares a balanced logistic-regression pipeline with a 300-tree balanced-subsample Random Forest. The saved baseline exposes eight features, uses threshold 0.31 selected by validation F1, and is loaded at FastAPI startup. For live Wokwi input, the four unavailable contextual fields are imputed with training-partition medians, while time-derived features are constructed from the timestamp. The 2026-10-11 direct prediction verifies loading, feature preparation, and response generation only; it is not a new accuracy, latency, generalisation, or complete telemetry-to-alert experiment.

Dataset source, licence, and label-generation provenance are not documented.

$$P_{avg} = (\frac{E_{tot}}{24D})\ x\ 1000\ W; \quad P_{base} = 0.20\ x\ P_{avg}$$

For threshold alerting, a scheduler evaluates the preceding one-hour household sum every 10 s. When alerting is enabled and the sum exceeds the user-defined threshold, an energy-alerts event is published. The Alert Service consumes threshold and ML anomaly paths, persists alerts in MySQL, and attempts SMTP delivery with Mailpit as the local demonstration inbox. Its deduplication key is userId:deviceId-or-HOUSEHOLD:alertType; matching keys are suppressed during the configured 15-min window. The active baseline ML worker includes a bounded default of three attempts and an anomaly-processing-failures topic as implemented handling, not as a separately evaluated recovery guarantee. The optional REFIT controlled endpoint and topic route remain isolated and disabled by default; they contributed no alert or empirical result in this paper.

TABLE I

Verified System Components, Current Baseline Flow, and Experimental Boundary

Stage	Implemented component and technique	Output
Access and telemetry	Keycloak JWT, roles, and ownership protect user routes; Wokwi-compatible ingestion uses a device-bound X-Device-Token; gateway and internal callers use separate trusted credentials.	Authorised user request or machine admission.
Ingestion and event transfer	Ingestion validates payloads and publishes Kafka energy-usage events. Usage and the ML worker consume the event independently.	Asynchronous event for persistence, aggregation, and baseline inference.
Storage and aggregation	Usage writes InfluxDB energy_usage points; User/Device metadata and alert records use MySQL.	Selected-window device and household totals.
Baseline model and insight	FastAPI hosts KafkaInferenceWorker and the fixed baseline Random Forest random_forest-csv-20261003T205417Z. Four absent Wokwi context fields are median-imputed; Insight functions are deterministic with optional Ollama fallback.	Anomaly probability/decision plus deterministic bill, ranking, standby, overview, and saving outputs.
Alerting, observation, and experimental boundary	Threshold and baseline anomaly paths reach Alert Service, which applies 15-min deduplication and uses Mailpit; Prometheus/Grafana observe services. REFIT controlled is separate and disabled.	Alert/email evidence, service metrics, and an explicitly inactive experimental branch.

C. Operational Workflow

Fig. 2 separates historical baseline artifact preparation from the current runtime telemetry path. The upper lane prepares the supplied CSV, selects the baseline Random Forest and threshold on validation data, and produces the fixed inference artifact. The active runtime lane keeps authorised telemetry admission, ingestion, Kafka publication, usage persistence, baseline feature construction and imputation, anomaly-result publication, deterministic insight generation, alert deduplication, and output delivery distinct. The dashed REFIT box is a separate disabled experimental route, not a deployed Wokwi or legacy-alert path. Historical E2E, security, deduplication, target-health, ingress-load, and selected outage checks are distinguished from the current 2026-10-11 availability, authorisation, event-path, and functional baseline-inference checks.

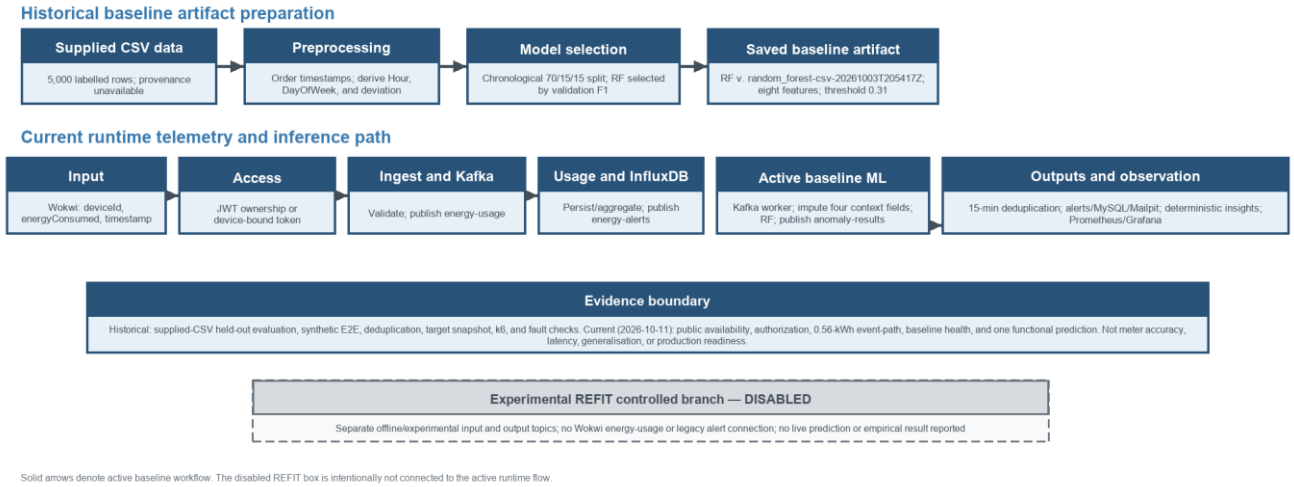


Fig. 2. Baseline artifact preparation, active telemetry workflow, and bounded historical/current verification evidence.

IV. IMPLEMENTATION AND FUNCTIONAL VERIFICATION

A. Implementation Environment

The system was exercised as a local, single-host Docker deployment on Windows 11 Home Single Language (10.0.26300) with eight logical processors and 7.74 GiB RAM. The compose configuration defines 17 services covering relational, event, time-series, e-mail, identity, application, frontend, and observability functions. On 2026-10-11, Gateway health, frontend, Keycloak OpenID Connect discovery, Prometheus readiness, and Grafana login each returned HTTP 200; these checks establish host availability, not production-scale operation. The environment supports controlled implementation and verification; it is not a replicated cloud or production deployment.

TABLE II

Local Experimental Environment and Evidence Conditions

Element	Documented configuration
Runtime and deployment	Windows 11 Home Single Language 10.0.26300; local single-host Docker with eight logical processors and 7.74 GiB RAM. The compose file defines 17 services. On 2026-10-11, Gateway, frontend, Keycloak OIDC, Prometheus readiness, and Grafana login each returned HTTP 200.
Application and data stack	Java 17 Spring Boot services and API Gateway; React/Vite frontend; FastAPI baseline ML service; Apache Kafka, MySQL, InfluxDB, Mailpit, Keycloak, Prometheus, and Grafana. The composition is locally deployed, not a cloud-scale result.
IoT simulation	Wokwi ESP32 DevKit with potentiometer, pushbuttons, and SSD1306 OLED. It posts Wokwi-compatible synthetic telemetry and is not a calibrated physical meter.
CSV data and baseline model	smart_meter_data.csv: 5,000 records, seven columns, and Normal/Abnormal labels of 4,750/250. A chronological 3,500/750/750 split selected the Random Forest. Artifact random_forest-csv-20261003T205417Z exposes eight features and threshold 0.31. Source, licence, and label provenance are undocumented.
Historical controlled local E2E	0.25, 0.50, and 0.75 kWh analytics inputs for user 15/device 314; a 1.00 kWh threshold input for user 16/device 315. These are prior synthetic REST observations, not a re-run after the current ML recovery.
Evaluation protocols	Historical held-out/model, E2E, deduplication, target-health, k6, and selected outage procedures; current 2026-10-11 availability, authorization/event-path, and direct baseline inference checks; separate 46-test Java, 18-test local Python, frontend lint, and build regressions.

B. Verification Procedure

1. **A prior controlled local E2E experiment submitted 0.25, 0.50, and 0.75 kWh analytics inputs for user 15/device 314 and a separate 1.00 kWh threshold input for user 16/device 315, whose configured threshold was 0.10 kWh. These historical synthetic REST observations were not silently re-presented as a post-recovery experiment.**
2. **The historical offline procedure stably sorted the supplied CSV, produced chronological train/validation/test partitions of 3,500/750/750 records, selected the threshold by validation F1, and evaluated the chosen and logistic-regression baseline models once on the held-out partition. The current baseline artifact metadata retains the same recorded model identity and threshold; its metrics are reported as historical artifact evaluation, not as a newly repeated live study.**
3. **On 2026-10-11, current runtime checks covered public availability, owned and cross-owner user access, device-bound telemetry admission, the Ingestion-to-Kafka-to-Usage-to-Influx path, protected insight endpoints, an administrator-only test alert, Mailpit capture, and temporary application-profile/device CRUD cleanup. A separate baseline ML health and direct prediction check established model loading, connected worker status, explicit imputation, and a response for a Wokwi-compatible payload; it did not rerun the complete ML-to-alert path after restart.**
4. **Historical local ingress was measured with k6 at 2, 5, and 10 requests/s for 30 s stages plus ramp-down; separate Mailpit and ML-service outage checks recorded ingress availability and manual container recovery. Current source-level regression evidence consists of four separately executed Java commands with 46 passing tests, an 18-test local Python ML suite, and successful frontend lint and production build. These execution records are reported separately and are not merged into one production-readiness score.**

V. RESULTS AND DISCUSSION

A. Functional Results

Table III separates historical offline and controlled results from current runtime verification because the measurements were collected by different procedures. Historical supplied-CSV assessment quantifies data-specific model performance, and historical E2E rows document synthetic integration conditions. Current 2026-10-11 evidence confirms availability, authorization/event-path behavior, and a live baseline endpoint/worker response, not a fresh statistical model evaluation or an ML-to-alert completion after restart. Source-level Java, Python, lint, and build records are likewise separate regression evidence rather than a unified production-readiness score.

TABLE III

Historical and Current Local Evaluation Evidence

Verification item	Condition	Observed result
Historical Random Forest held-out test	750 records; selected threshold = 0.31; artifact-evaluation condition.	TN 705, FP 5, FN 3, TP 37; precision 0.880952; recall 0.925000; F1 0.902439; ROC-AUC 0.997430; PR-AUC 0.948532.
Historical Logistic-Regression baseline	Same held-out partition; threshold = 0.28.	Precision 0.053333; recall 1.000000; F1 0.101266; ROC-AUC 0.422852; PR-AUC 0.059710.
Historical controlled E2E analytics	Three synthetic inputs total 1.50 kWh.	3/3 HTTP 201; cost $0.18; daily 0.50 kWh; projected 15.00 kWh/$11.80; rank 100%; standby 4.17 W/LOW; observed within polling-bounded 3.79 s.
Historical threshold alert and Mailpit	1.00 kWh input exceeds 0.10 kWh threshold.	HTTP 201; persisted sent THRESHOLD alert and matching Mailpit email; observed within polling-bounded 13.64 s.
Historical gateway security	Eight configured access-control cases.	8/8 expected statuses: owned 200; missing/invalid/expired JWT 401; cross-user 403; administrator 200; device-token ingestion 201.
Historical alert deduplication	Two identical 1.00-kWh events at the same timestamp.	One ML_ANOMALY alert was stored and a duplicate was explicitly suppressed within 15 min.
Historical observability snapshot; current host availability	Historical Prometheus target snapshot; current readiness/login routes.	Historical 8/8 active targets up. Current Prometheus readiness and Grafana login returned HTTP 200; no new all-target snapshot is claimed after ML restart.
Historical local ingress and fault checks	k6 2/5/10 requests/s; Mailpit and ML separately stopped.	450/450 accepted; mean 39.442 ms; p95 109.773 ms; 0% HTTP failure. During each outage, ingestion returned 201 and manual restart recovered the container.
Current active baseline ML runtime (2026-10-11)	Live /health: baseline UP, legacy Kafka worker connected, REFIT controlled DISABLED. Direct Wokwi-compatible input: deviceId 1, energyConsumed 0.55, timestamp.	anomaly = false; anomalyProbability = 0.0; decisionThreshold = 0.31; four imputed features: Temperature, Humidity, Wind_Speed, Avg_Past_Consumption. Functional check only.
Current authorization and event-path validation (2026-10-11)	Five user-route checks; five telemetry-admission checks; accepted readings 0.37 and 0.19 kWh; protected insights and administrator alert checked.	User checks: 401/200/403/403/200; telemetry: 401/401/201/201/403; Usage changed 0.00 to 0.56 kWh (>=0.559 lower bound). Alert ID 492 reported sent = true and Mailpit API returned 200.

B. Interpretation and Evidence Boundaries

On the supplied held-out CSV, the Random Forest achieved substantially stronger precision, F1, ROC-AUC, and PR-AUC than the logistic-regression baseline. This supports selection of the saved Random Forest artifact for this dataset and split, but it does not validate field anomaly detection. The current direct Wokwi-compatible prediction confirms that the loaded baseline can construct features, explicitly impute missing context, and return a decision; it does not add an accuracy observation or prove the post-restart telemetry-to-ML-to-alert route. The historical E2E analytics and alert observations demonstrate a traceable local integration path from synthetic REST input to stored usage, deterministic outputs, and a captured notification; they do not establish physical-meter accuracy or utility-bill agreement.

The historical gateway-status, deduplication, Prometheus target, k6, and selected outage records remain bounded to their stated dates and procedures. The current post-hardening run adds five user-route checks, five telemetry-admission checks, an observed 0.00-to-0.56 kWh Usage change after 0.37 and 0.19 kWh readings, protected insight responses, one administrator-only alert with Mailpit capture, and temporary CRUD cleanup. Static gateway and internal-service credentials provide defence in depth, but they are not mTLS, workload identity, secret rotation, replay protection, or penetration-test evidence. No fresh all-target Prometheus snapshot was captured after the ML service restart.

The data and runtime boundaries remain material. The CSV has no documented source, licence, label provenance, household/device identifier, or provenance for Avg_Past_Consumption. Its Electricity_Consumed values are in an undocumented 0-1 range, whereas Wokwi modes send direct values in approximately the 1-40 range; optional Wokwi context is median-imputed and no range-validation or calibration experiment is reported. The REFIT controlled branch is disabled because its optional artifacts are unavailable to the current Docker build and contributes no result. No runtime inference-latency distribution was captured; the current single endpoint invocation is a functional check rather than a latency benchmark. Frontend rendering alone was not used as backend-data evidence

because local fallback data may be displayed if an API request fails.

VI. LIMITATIONS AND FUTURE WORK

The principal limitations are synthetic, non-calibrated Wokwi telemetry; the anonymous dataset and missing provenance; the training-to-Wokwi unit/range mismatch; and the absence of physical-meter, utility-bill, and external-field validation. The optional REFIT controlled branch is disabled and lacks a reported live or real-anomaly evaluation. The evaluation uses one local Docker host with a bounded ingress test, no distributed tracing or Kafka-lag study, and only selected manually recovered service-outage checks. Keycloak, gateway, device-bound token, and downstream-caller controls were functionally checked, but the local development deployment uses static shared credentials rather than TLS, mTLS, workload identity, credential rotation, replay protection, or penetration testing. No browser role/refresh acceptance test and no fresh post-restart all-target Prometheus snapshot were captured.

Future work should collect consented, documented, calibrated interval-meter data with per-device and household context; establish label and feature provenance; and add range or out-of-distribution checks before model deployment. It should make the experimental REFIT artifacts accessible only after an audited dataset, label process, and isolated evaluation protocol are available, then compare it without transferring baseline metrics. Further work should repeat end-to-end and load measurements over longer workloads, add tracing and Kafka-lag instrumentation, and perform automated fault, retry, failure-topic, replay, and recovery experiments. Production hardening should address TLS, rotating secrets or workload identity, device provisioning, browser acceptance tests, and an independent security assessment before any real-world deployment claim.

VII. CONCLUSION

This paper presents a microservices-based Home Energy Tracker updated from an integration baseline to a locally evaluated event-driven system with gateway access controls, Kafka-mediated processing, InfluxDB and MySQL persistence, deterministic insights, a deployed baseline Random Forest inference artifact, alert deduplication, and Prometheus/Grafana observation. Historical evidence includes the supplied-CSV model assessment, synthetic E2E analytics and alert checks, gateway-security cases, duplicate suppression, an eight-target snapshot, local k6 ingress measurement, and two limited outage-availability observations. Current 2026-10-11 verification additionally shows public service availability, access-controlled event propagation, a live baseline ML container and connected worker, and one Wokwi-compatible prediction with explicit imputation; the REFIT controlled branch remained disabled. The work demonstrates implementation-level integration and bounded local verification; it does not establish meter accuracy, externally generalisable anomaly detection, production-scale performance, comprehensive resilience, or production security.

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